# Two-stage Gauss–Legendre Method

Exact one-step Jacobian, symplecticity, and symmetry

Fourth-order collocation method for canonical Hamiltonian systems

## 1 Hamiltonian convention

Let $z \in \mathbb{R}^{2d}$ and consider the possibly time-dependent canonical Hamiltonian system

$$\dot{z} = f(t,z) = \Omega^{-1}\nabla_z H(t,z), \qquad \Omega = \begin{pmatrix} 0 & I_d \\ -I_d & 0 \end{pmatrix}. \tag{1}$$

For the component-major guiding-centre state

$$z = (x_1, \dots, x_d, y_1, \dots, y_d)^T,$$

this convention gives

$$f(t,z) = \begin{pmatrix} -\nabla_y H(t,z) \\ \nabla_x H(t,z) \end{pmatrix}.$$

At any fixed time and state, let $F = D_z f(t,z)$. Since $\nabla_z^2 H$ is symmetric,

$$F^T\Omega + \Omega F = 0. \tag{2}$$

This identity is the infinitesimal Hamiltonian condition used below. The proof remains valid for a time-dependent Hamiltonian because the stage times do not depend on the initial state $z_n$.

## 2 Gauss–Legendre stages

Set

$$r = \frac{\sqrt{3}}{6}, \qquad c_1 = \frac{1}{2} - r, \qquad c_2 = \frac{1}{2} + r.$$

The Butcher coefficients of the two-stage Gauss–Legendre method are

$$A = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} - r \\ \frac{1}{4} + r & \frac{1}{4} \end{pmatrix}, \qquad b = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}. \tag{3}$$

For one step of length $h$, the stage states satisfy

$$Z_i = z_n + h\sum_{j=1}^{2} a_{ij} f(t_n + c_j h, Z_j), \qquad i = 1,2, \tag{4}$$

and the accepted state is

$$z_{n+1} = \Phi_{h,t_n}(z_n) = z_n + h\sum_{i=1}^{2} b_i f(t_n + c_i h, Z_i). \tag{5}$$

The method is implicit: equations (4) determine $(Z_1, Z_2)$ simultaneously.

## 3 Direct proof of fourth-order accuracy

We now prove directly why the Runge–Kutta order conditions through degree four give a fourth-order method. For clarity, first consider the autonomous problem

$$\dot{y} = f(y), \qquad y(t_n) = y. \tag{6}$$

The non-autonomous case follows by augmenting the state with time, $\widehat{y} = (t, y)$ and $\dot{\widehat{y}} = (1, f(t, y))$. Because $c = A\mathbf{1}$, applying the Runge–Kutta method to this augmented autonomous system produces exactly the stage times used in (4).

Set

$$F = f(y), \qquad F' = Df(y), \qquad F'' = D^2 f(y), \qquad F''' = D^3 f(y), \tag{7}$$

where the derivatives are multilinear maps. Repeated application of the chain rule gives

$$\begin{aligned} y' &= F, \\ y'' &= F'F, \\ y''' &= F''(F,F) + F'^2 F, \\ y^{(4)} &= F'''(F,F,F) + 3F''(F,F'F) + F'F''(F,F) + F'^3 F. \end{aligned} \tag{8}$$

Consequently, the exact flow has the expansion

$$\begin{aligned} y(t_n + h) = y + hF &+ \frac{h^2}{2} F'F \\ &+ \frac{h^3}{6}\left[F''(F,F) + F'^2 F\right] \\ &+ \frac{h^4}{24}\left[F'''(F,F,F) + 3F''(F,F'F) + F'F''(F,F) + F'^3 F\right] + O(h^5). \end{aligned} \tag{9}$$

For a generic Runge–Kutta step, write

$$K_i = f(Y_i), \qquad Y_i = y + h\sum_j a_{ij} K_j, \qquad y_{n+1} = y + h\sum_i b_i K_i, \tag{10}$$

and define $C = \mathrm{diag}(c)$. Powers such as $c^2$ and $c^3$ are understood componentwise. If $\Delta_i = h\sum_j a_{ij} K_j$, Taylor expansion about $y$ gives

$$K_i = F + F'\Delta_i + \frac{1}{2}F''(\Delta_i, \Delta_i) + \frac{1}{6}F'''(\Delta_i, \Delta_i, \Delta_i) + O(h^4). \tag{11}$$

Substituting the stage expansions recursively into (11) and collecting equal powers of $h$ yields

$$\begin{aligned} K_i = F &+ hc_i F'F \\ &+ h^2\left[(Ac)_i F'^2 F + \frac{1}{2}c_i^2 F''(F,F)\right] \\ &+ h^3\left[(A^2 c)_i F'^3 F + \frac{1}{2}(Ac^2)_i F'F''(F,F)\right. \\ &\qquad \left. + c_i (Ac)_i F''(F,F'F) + \frac{1}{6}c_i^3 F'''(F,F,F)\right] + O(h^4). \end{aligned} \tag{12}$$

Inserting (12) into the accepted update in (10) gives

$$\begin{aligned} y_{n+1} = y &+ h(b^T\mathbf{1})F + h^2(b^T c)F'F \\ &+ h^3\left[(b^T Ac)F'^2 F + \frac{1}{2}(b^T c^2)F''(F,F)\right] \end{aligned}$$

$$+ h^4 \Big[ (b^T A^2 c) F'^3 F + \frac{1}{2}(b^T A c^2) F' F''(F,F) + (b^T C A c) F''(F, F'F) + \frac{1}{6}(b^T c^3) F'''(F,F,F) \Big] + O(h^5). \tag{13}$$

The elementary differentials in (9) and (13) are independent for a general smooth vector field. Their coefficients agree through $h^4$ precisely when

$$\begin{aligned} b^T \mathbf{1} &= 1, & b^T c &= \frac{1}{2}, \\ b^T c^2 &= \frac{1}{3}, & b^T A c &= \frac{1}{6}, \\ b^T c^3 &= \frac{1}{4}, & b^T C A c &= \frac{1}{8}, \\ b^T A c^2 &= \frac{1}{12}, & b^T A^2 c &= \frac{1}{24}. \end{aligned} \tag{14}$$

For example, the coefficient of $F''(F,F)$ in the numerical step is $\frac{1}{2} b^T c^2$ and its exact coefficient is $\frac{1}{6}$, which gives $b^T c^2 = \frac{1}{3}$. The other seven equalities follow from the same term-by-term comparison. Thus these conditions are not merely formal identities: they are exactly the equations that make the numerical and exact Taylor expansions coincide through fourth degree.

It remains to verify them for the coefficients in (3). Direct calculation gives

$$b^T \mathbf{1} = 1, \qquad b^T c = \frac{1}{2}, \qquad b^T c^2 = \frac{1}{3}, \qquad b^T c^3 = \frac{1}{4}, \tag{15}$$

and

$$Ac = \frac{1}{2} c^2, \qquad b^T A = b^T (I - C). \tag{16}$$

Therefore

$$\begin{aligned} b^T A c &= \frac{1}{2} b^T c^2 = \frac{1}{6}, & b^T C A c &= \frac{1}{2} b^T c^3 = \frac{1}{8}, \\ b^T A c^2 &= b^T (c^2 - c^3) = \frac{1}{12}, & b^T A^2 c &= \frac{1}{2} b^T A c^2 = \frac{1}{24}. \end{aligned} \tag{17}$$

Hence all conditions in (14) hold, and comparison of (9) with (13) proves the one-step defect estimate

$$\| y(t_n + h) - y_{n+1} \| = O(h^5). \tag{18}$$

Finally, let $e_n$ be the global error. For a smooth Lipschitz vector field, the Runge–Kutta one-step map is Lipschitz with factor $1 + Lh$ for sufficiently small $h$. The local estimate (18) then gives

$$\| e_{n+1} \| \le (1 + Lh) \| e_n \| + C h^5. \tag{19}$$

For $nh \le T$, the discrete Gronwall estimate yields

$$\| e_n \| \le C h^5 \sum_{j=0}^{n-1} (1 + Lh)^j \le C_T h^4. \tag{20}$$

Thus the two-stage Gauss–Legendre method has global order four.

## 4 Exact Jacobian of the stage root

Introduce the stage field Jacobians and stage sensitivities

$$F_i := D_z f(t_n + c_i h, Z_i), \qquad S_i := \frac{\partial Z_i}{\partial z_n}. \tag{21}$$

Differentiating (4) with respect to $z_n$ gives

$$S_i = I_{2d} + h \sum_{j=1}^{2} a_{ij} F_j S_j. \tag{22}$$

Equivalently,

$$\underbrace{\begin{pmatrix} I_{2d} - \frac{h}{4} F_1 & -h\left(\frac{1}{4} - r\right) F_2 \\ -h\left(\frac{1}{4} + r\right) F_1 & I_{2d} - \frac{h}{4} F_2 \end{pmatrix}}_{M = D_{(Z_1, Z_2)} R} \begin{pmatrix} S_1 \\ S_2 \end{pmatrix} = \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}. \tag{23}$$

Here $M$ is the Jacobian of the coupled stage residual. It is also the matrix used by an exact Newton linearization, but it is *not* the Jacobian of the physical one-step map.

Assuming that the converged stage root is regular, the implicit function theorem gives

$$\begin{pmatrix} S_1 \\ S_2 \end{pmatrix} = M^{-1} \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}. \tag{24}$$

In an implementation, equation (23) should be solved directly; the inverse in (24) is only notation.

## 5 Exact Jacobian of the physical step

Define the Jacobian of the map (5) by

$$\mathcal{J}_n := D_{z_n} \Phi_{h, t_n}(z_n).$$

Differentiating the accepted update and using (21) yields

$$\boxed{\mathcal{J}_n = I_{2d} + h \sum_{i=1}^{2} b_i F_i S_i = I_{2d} + \frac{h}{2} \left(F_1 S_1 + F_2 S_2\right).} \tag{25}$$

Combining (24) and (25), the same Jacobian can be written as

$$\boxed{\mathcal{J}_n = I_{2d} + h \begin{pmatrix} \frac{1}{2} F_1 & \frac{1}{2} F_2 \end{pmatrix} M^{-1} \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}.} \tag{26}$$

This is the exact tangent of the ideal converged root. It differentiates the mathematical Gauss–Legendre map, not a finite Newton stopping rule.

## 6 Direct proof of symplecticity with the step Jacobian

The one-step map is symplectic if and only if

$$\mathcal{J}_n^T \Omega \mathcal{J}_n = \Omega. \tag{27}$$

Set

$$K_i := F_i S_i. \tag{28}$$

Then equations (25) and (22) become

$$\mathcal{J}_n = I_{2d} + h \sum_i b_i K_i, \qquad I_{2d} = S_i - h \sum_j a_{ij} K_j. \tag{29}$$

Expanding the symplecticity defect gives

$$\begin{aligned} \mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega &= h \sum_i b_i \left( K_i^T \Omega + \Omega K_i \right) \\ &\quad + h^2 \sum_{i,j} b_i b_j K_i^T \Omega K_j. \end{aligned} \tag{30}$$

By the Hamiltonian identity (2),

$$\begin{aligned} K_i^T \Omega S_i + S_i^T \Omega K_i &= S_i^T \left( F_i^T \Omega + \Omega F_i \right) S_i \\ &= 0. \end{aligned} \tag{31}$$

Using the second identity in (29) on both occurrences of $I_{2d}$ therefore gives

$$K_i^T \Omega + \Omega K_i = -h \sum_j a_{ij} \left( K_i^T \Omega K_j + K_j^T \Omega K_i \right). \tag{32}$$

Substitute (32) into (30). Renaming $i$ and $j$ in the second term arising from (32) produces

$$\boxed{\mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega = h^2 \sum_{i,j} \left( b_i b_j - b_i a_{ij} - b_j a_{ji} \right) K_i^T \Omega K_j.} \tag{33}$$

Consequently, the defect vanishes whenever the Runge–Kutta coefficients obey

$$\boxed{b_i a_{ij} + b_j a_{ji} = b_i b_j, \qquad i, j = 1, 2.} \tag{34}$$

# 7 Verification for the Gauss–Legendre coefficients

For the diagonal terms in (34),

$$2b_1a_{11} - b_1^2 = 2\left(\frac{1}{2}\right)\left(\frac{1}{4}\right) - \frac{1}{4} = 0, \qquad 2b_2a_{22} - b_2^2 = 0. \tag{35}$$

For the off-diagonal terms,

$$\begin{aligned} b_1a_{12} + b_2a_{21} - b_1b_2 &= \frac{1}{2}\left(\frac{1}{4} - r\right) + \frac{1}{2}\left(\frac{1}{4} + r\right) - \frac{1}{4} \\ &= 0. \end{aligned} \tag{36}$$

The case $(i, j) = (2, 1)$ is the same equality with the two summands exchanged. Thus every coefficient multiplying $K_i^T \Omega K_j$ in (33) is zero, and hence

$$\boxed{\mathcal{J}_n^T \Omega \mathcal{J}_n = \Omega.} \tag{37}$$

Therefore the two-stage Gauss–Legendre method is symplectic when its implicit stage equations are solved exactly.

# 8 Symmetry and time reversibility

Symmetry is a property distinct from symplecticity. The method is symmetric when its backward step is exactly the inverse of its forward step. For the possibly time-dependent problem (1), this means

$$\boxed{\Phi_{-h,t_n+h} \circ \Phi_{h,t_n} = \mathrm{Id}\,.} \tag{38}$$

For an autonomous problem, equation (38) reduces to $\Phi_{-h} = \Phi_h^{-1}$.

For the forward step, write

$$f_1 = f(t_n + c_1 h, Z_1), \qquad f_2 = f(t_n + c_2 h, Z_2). \tag{39}$$

Equations (4) and (5) are then

$$\begin{aligned} Z_1 &= z_n + h(a_{11} f_1 + a_{12} f_2), \\ Z_2 &= z_n + h(a_{21} f_1 + a_{22} f_2), \qquad (40) \\ z_{n+1} &= z_n + h(b_1 f_1 + b_2 f_2). \qquad (41) \end{aligned}$$

Now start from $z_{n+1}$ at time $t_n + h$ and apply the same method with step $-h$. Its stages $\widehat{Z}_1, \widehat{Z}_2$ satisfy

$$\widehat{Z}_i = z_{n+1} - h \sum_{j=1}^{2} a_{ij} f(t_n + h - c_j h, \widehat{Z}_j). \tag{42}$$

The Gauss nodes are reflected about the midpoint:

$$1 - c_1 = c_2, \qquad 1 - c_2 = c_1. \tag{43}$$

Thus the backward stage times are the forward stage times in reverse order. We shall verify that the backward stages are

$$\boxed{\widehat{Z}_1 = Z_2, \qquad \widehat{Z}_2 = Z_1.} \tag{44}$$

Under the identification (44), the first backward stage equation becomes

$$\begin{aligned} \widehat{Z}_1 &= z_{n+1} - h(a_{11} f_2 + a_{12} f_1) \\ &= z_n + h\left[(b_1 - a_{12}) f_1 + (b_2 - a_{11}) f_2\right]. \end{aligned} \tag{45}$$

For the coefficients (3),

$$b_1 - a_{12} = a_{21}, \qquad b_2 - a_{11} = a_{22}. \tag{46}$$

Consequently,

$$\widehat{Z}_1 = z_n + h(a_{21} f_1 + a_{22} f_2) = Z_2.$$

Likewise, the second backward stage satisfies

$$\begin{aligned} \widehat{Z}_2 &= z_{n+1} - h(a_{21} f_2 + a_{22} f_1) \\ &= z_n + h\left[(b_1 - a_{22}) f_1 + (b_2 - a_{21}) f_2\right]. \end{aligned} \tag{47}$$

Since

$$b_1 - a_{22} = a_{11}, \qquad b_2 - a_{21} = a_{12}, \tag{48}$$

we obtain

$$\widehat{Z}_2 = z_n + h(a_{11} f_1 + a_{12} f_2) = Z_1.$$

The proposed reversed stages therefore solve the coupled backward stage equations exactly.

The output of the backward step is now

$$\begin{aligned}\widehat{z}_n &= z_{n+1} - h(b_1 f_2 + b_2 f_1) \\ &= z_n + h(b_1 f_1 + b_2 f_2) - h(b_1 f_2 + b_2 f_1) \\ &= z_n, \end{aligned} \tag{49}$$

where the last equality uses $b_1 = b_2 = 1/2$. This proves (38).

More generally, an $s$-stage Runge–Kutta method is symmetric when its coefficients are invariant under stage reversal:

$$\boxed{c_i = 1 - c_{s+1-i}, \qquad b_i = b_{s+1-i}, \qquad a_{ij} = b_{s+1-j} - a_{s+1-i,s+1-j}.} \tag{50}$$

The coefficients in (3) satisfy all three identities for $s = 2$.

Differentiating the map identity (38) also gives the Jacobian form of reversibility:

$$\boxed{D\Phi_{-h,t_n+h}(z_{n+1})\, D\Phi_{h,t_n}(z_n) = I_{2d}.} \tag{51}$$

Thus the backward-step Jacobian is exactly the inverse of the forward-step Jacobian. As with symplecticity, a finite Newton tolerance and floating-point round-off can produce a small measured reversibility defect even though the ideal implicit method is exactly symmetric.

# 9 Implemented configurations

The public `GaussLegendre4` model uses the same collocation equations for every configuration. Configure differentiation with `solver=NewtonSolver(jacobian_method=...)`: the accepted values are `analytic`, `finite_difference`, and `auto`. Execution supplies the physical vector field and optional Jacobian to the method. Analytic mode uses independent particle-block derivatives; finite differences approximate the stage-field derivatives before assembling the coupled Newton system. Auto selects the supplied derivative when available and otherwise uses finite differences. Absolute and relative tolerances and the maximum iteration count also belong to the configured solver.

The generic `EnergyObserver` evaluates physical $H$ and analytic positive $\partial_t H$ at accepted endpoints and requested physical samples. It records energy differences without integrating momentum or using internal stages. A historical $H + \kappa$ study must explicitly request the separate stage-based diagnostic. Physical trajectories and solver equations are unchanged.

# 10 Consequences and numerical interpretation

For one guiding-centre particle, $\mathcal{J}_n \in \mathbb{R}^{2\times 2}$ and

$$\mathcal{J}_n^T \Omega \mathcal{J}_n = \det(\mathcal{J}_n)\Omega.$$

Equation (37) therefore implies

$$\det(\mathcal{J}_n) = 1,$$

so the one-step map preserves oriented area. In dimensions greater than two, unit determinant is only a consequence of symplecticity, not an equivalent test.

If $\mathcal{J}_n$ denotes the local Jacobian at step $n$, the accumulated flow tangent after $N$ steps is

$$\mathcal{P}_N = \mathcal{J}_{N-1} \cdots \mathcal{J}_1 \mathcal{J}_0.$$

Since a product of symplectic matrices is symplectic,

$$\mathcal{P}_N^T \Omega \mathcal{P}_N = \Omega.$$

The algebraic result concerns the exact root and the exact tangent (25). In floating-point computations, a finite Newton tolerance, round-off, or a finite-difference approximation of the map Jacobian produces a small nonzero measured defect

$$\left\| \mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega \right\|.$$

That numerical floor does not contradict the exact symplecticity proved above.

# References


- J. M. Sanz-Serna, “Runge–Kutta schemes for Hamiltonian systems,” *BIT Numerical Mathematics*, 28 (1988), 877–883.
- E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, second edition, Springer, 2006.

# 11 Physical state and endpoint energy observations

Accepted states contain one physical vector $Z \in \mathbb{R}^{dN}$ and scalar time. GC vectors have $2N$ entries and FC vectors have $4N$; doubled workspaces are transient. InitialConditions supplies values and layout during preparation only. The numerical base retains functions, metadata, run status and copy count; execution owns the initial snapshot and output schedule. Supplied Jacobians are independent particle blocks $(N, d, d)$; finite differences use dense matrices.

The generic `EnergyObserver` evaluates only

$$H_n = H(t_n, Z_n), \quad H_{n+1} = H(t_{n+1}, Z_{n+1}), \quad (\partial_t H)_n, \quad (\partial_t H)_{n+1}, \quad \Delta H_n = H_{n+1} - H_n.$$

It also evaluates these physical functions at requested saved samples, without consuming internal stages or integrating a momentum coordinate. The derivative is positive, analytic and partial at fixed state. GC differentiates its normalized Hamiltonian potential with `dt=1`; spatial splines and Fourier time factors retain their original scaling. This is not the secant $\Delta H_n/h$. Variation of a time-dependent Hamiltonian is not a numerical conservation error.

Specialized historical generalized-energy studies separately reconstruct $\dot{\kappa} = -\partial_t H$ from retained numerical stages. Their signed weights, physical normalization and archived diagnostics keep their original meaning. They are not part of generic energy observation. Host and JAX execution share the fixed numerical map and physical-only accepted state. JAX evaluates endpoint energy on the device without a momentum accumulator.

# 12 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Newton systems use the supplied particle-block structure. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.